AI Applications in Animal Health

A Critical Literature Review of State-of-the-Art Methods

Vaxxinova | Disease Forecasting Initiative

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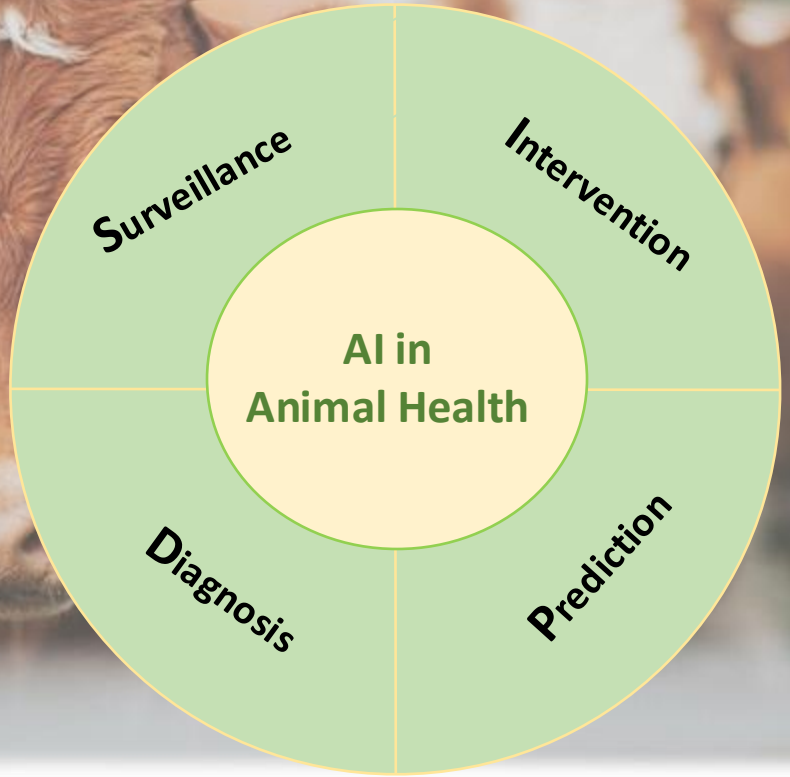
1. Landscape


Surveillance
Intervention
AI in
Animal Health
Diagnosis
Prediction

Overview of AI Applications in Animal Health

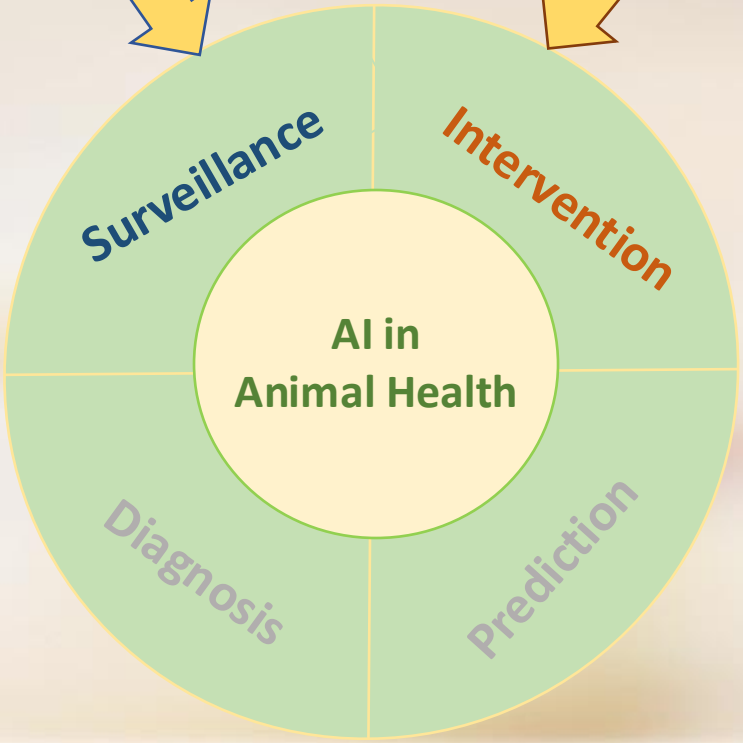
Ye et al. (2025) — Nature Communications; Guo et al. (2023) — Sci. In One Health

AI Application	Known Disease (examples)	Unknown / Emerging Disease (examples)	Pattern Recognition	Prediction
Surveillance	Early infection detection	Early (infection detection + anomaly detection)	✓	✗
Intervention	Control strategies	Vaccine design (e.g., antigen selection)	✓	✗
Diagnosis	Infection identification	Novel pathogen identification	✓	✗
Prediction	Outbreak trajectory	*Limited (insufficient data)*	✓	✓

AI Capabilities :

Pattern recognition (what is this? / what happened?) | Prediction (what will happen next?)

2. Examples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AI in Surveillance & Intervention

Current Literature Status:
Surveillance (**extensive & quality vary**)
Vaccine Design (**emerging & challenging**)

E1

Surveillance: Ill Pig Early Detection [Jorquera-Chavez et al. (2021)-Computers and Electronics in Agriculture]

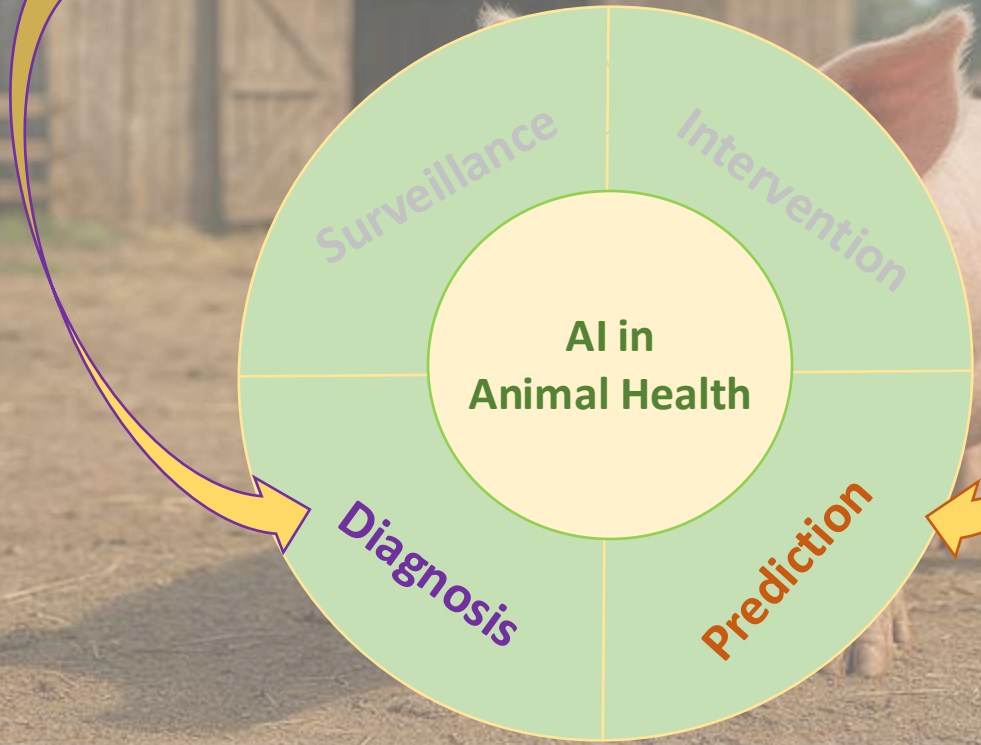
Input: sensor (video + thermal sensors)
Output: sick/healthy pig classification before clinical sign (respiratory related diseases)

E2

Vaccine Design: Identifying vaccine targets for new pathogens [Villanueva-Flores et al. (2025)-npj Vaccines]

Input: sequence (e.g., motifs) + structure (e.g., surface accessibility) of antigen (determines vaccine's success)
Output: vaccine target (pursue) or not (skip)

3. Examples



Methods:

- *Pure AI*
- *Hybrid AI*

Pure AI : quick prediction / no reasoning

Trend Prediction: PEDV in Pigs [Ajayi et al. (2019)-Preventive Veterinary Medicine]

Input: weather (e.g., temperature) + disease records (e.g., prevalence)
Output: number of new cases over next 4 weeks

Pros

- Fast development (existing AI + data)
- Handles complex, messy data (image, text, audio, number)
- No prior knowledge of domain knowledge (e.g., disease spread or biology) required

Cons

- Heavy reliance on data quality and quantity
- Black box: provides output only
  e.g., 4 new cases in next 4 weeks, no why or what factors drive it

Hybrid AI (AI+Epi.): prediction + understanding

Methods: Combines AI analysis with disease transmission science to forecast outbreaks

Trend forecasting: BRD in Cattle [Tabi et al. (2024) — Preventive Veterinary Medicine]

AI model
diagnostics

Input:
lung ultrasound videos

Output: # of infections

→

Bridge
parameter estimation

e.g., transmission rate,
initial prevalence

→

Epi. Model
disease transmission

Output:
infection count every 12 hrs / 30 days

Hybrid (AI+Epi.): prediction + understanding

Pros

- Explanatory prediction: can trace the reasoning behind the prediction outcome
  e.g., infections detected → transmission rate estimated → forecast: X infected by D7, Y by D14
- Can perform better than pure AI when data is incomplete or noisy
- Can simulate "what if" scenarios to guide intervention decisions
  e.g., impact of reducing herd density by 20% vs. 40% on disease outbreak

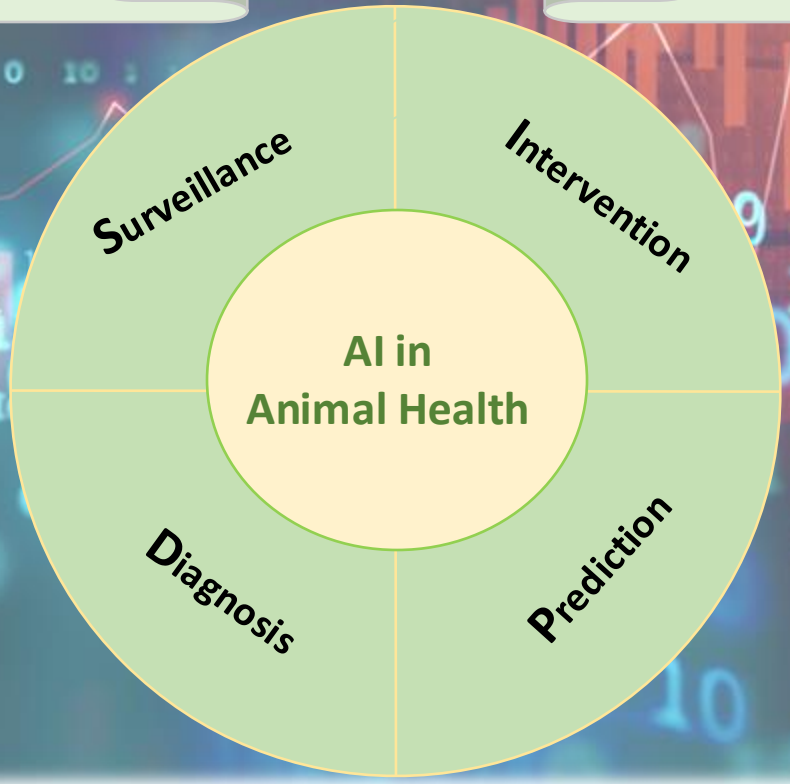
Cons

- Longer development: AI experts + epidemiology experts
- Model complexity: each model carries its own limitations — poor integration doubles the risk

Comparison (predicting)

	Pure AI	Hybrid (AI+Epi.)
Overview	Quick result / No need reasoning	Prediction + Understanding
Best for	Short-term forecasting	Long-term planning & scenario analysis
Explains forecast reasoning	✗	✓
Intervention testing	✗	✓
Development	Fast	Slow
Complexity	AI model selection	AI + epidemiology model selection & integration
Literature evidence	Well-established in animal health	Well-established in human disease, emerging in animal health

4. Data for AI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Key Data Inputs

Heinen, Larson & White (2025) — Animals (Basel), Ezanno et al. (2021)-Vet Res.

Data type	Example
Disease records	Historical outbreak records (time, location, prevalence)
Physiological features	Body temperature, heart rate
Demographic data	Animal, group, and farm-level data (density, farm type, feeding-related)
Laboratory data	Pathogen sequences, PCR
Sensor data	Audio (coughing), video (movement), ultrasound
Climate data	Temperature, humidity
Social media & news	Outbreak alerts (location, severity)

Multi-source Data Collection Trade-off:
Benefit: comprehensive information & enhanced accuracy
Risk: costly and data quality varies across sources

Literature suggests: Start with readily available data (disease records, climate data) before expanding to costlier sources (sensors, lab data)

Topic Discussion

- AI in Animal Health-- Challenges & Actions
- Ming's Suggestions

AI in Animal Health-- Challenges & Actions

Challenges:

- **Data collection:** non-standardized, diverse, security & privacy constraints
- **Limited practical validation:** small, localized case studies (e.g., 170 pigs in PEDV, 163 cows in BRD)
- **Cross-disciplinary** collaboration needed

Actions:

- DECIDE : AI-driven decision support for animal disease control by 20 institutions of EU *(research stage)*
- CatGPT: AI chatbot by Cornell College of Veterinary Medicine for cat health queries
- Animal Disease Insights: dashboard tracking disease outbreaks worldwide using WOAH data

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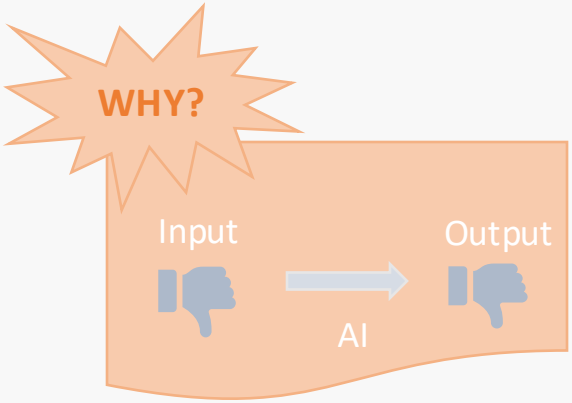
Ming's Suggestions

Involve domain experts (e.g., vets, farmers):

- **Before data collection** (heuristic start)
  - Choose relevant factors
  - Select data horizon (based on disease cycle frequency)
- **Before modeling** *(validation)*
  - Validate data values make biological sense
- **After modeling** *(validation)*
  - Verify predictions/identifications make practical sense



Selected References

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2. Jorquera-Chavez et al. (2021): Using imagery and computer vision as remote monitoring methods for early detection of respiratory disease in pigs - ScienceDirect
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7. Heinen, Larson & White (2025) — Animals (Basel):Disease Prediction in Cattle: A Mixed-Methods Review of Predictive Modeling Studies - PubMed
8. Villanueva-Flores et al. (2025)-npj Vaccines: AI-driven epitope prediction: a systematic review, comparative analysis, and practical guide for vaccine development | npj Vaccines